The Invariance Principle — problem 1.20

A. Engel — Problem-Solving Strategies · 1998

A game for computing $\gcd(a, b)$ and $\text{lcm}(a, b)$.
We start with $x = a$, $y = b$, $u = a$, $v = b$ and move as follows :
if $x < y$ then, set $y \leftarrow y - x$ and $v \leftarrow v + u$
if $x > y$, then set $x \leftarrow x - y$ and $u \leftarrow u + v$
The game ends with $x = y = \gcd(a, b)$ and $(u + v)/2 = \text{lcm}(a, b)$. Show this.